

Collins Anyanwu

Senior Geodatabase & GIS Engineer | Product Engineering • SQL Server, Oracle & PostGIS • ArcGIS Enterprise

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Professional Summary

Enterprise geodatabase specialist with strong DBMS skills across Microsoft SQL Server, Oracle Spatial, and PostgreSQL/PostGIS, implementing ArcGIS Pro, ArcMap, and ArcGIS Enterprise solutions. Prepares and maintains the electric utility network model that feeds Ikeja Electric's ADMS, validates connectivity and topology, produces network diagrams (schematics), and was part of the ADMS rollout; gathers user requirements, designs data models, manages geodatabase versioning and replication, publishes data, map, geodata and OGC (WFS) services to ArcGIS Enterprise and Online, and supports users. Boosted database performance by 50%, uses partitioning, automates with Python/ArcPy and FME, and builds versioned FastAPI REST/JSON APIs, with test plans, automated and regression testing, release work, and AI tools (LLMs, GeoAI).

Core Skills

Enterprise Geodatabase Administration (SQL Server, Oracle Spatial, PostgreSQL/PostGIS, ArcSDE) • Geodatabase Versioning & Replication • Multi-User Access Control (RBAC, SSO) • Database Design, Partitioning, Query Tuning & Migration • Geodatabase Data Models & Data Standards • Topology, Connectivity & Network Diagrams (Schematics) • Utility Network Data for ADMS • Publishing Data & Services (ArcGIS Enterprise, ArcGIS Online, Geodata Services, OGC WFS) • ArcGIS Pro, ArcMap & ArcGIS Enterprise • Python/ArcPy & FME Automation • Versioned REST/JSON APIs (FastAPI) • Test Plans, Automated, Functional & Regression Testing & Release • Data Quality Assurance & Validation • User Requirements Gathering • Technical Documentation • GIS User Support & Troubleshooting • Server Administration (PostGIS & AWS EC2) • Project Delivery & Team Leadership • GeoAI (ML & Deep Learning), LLMs, Prompt Engineering & AI Coding Tools • Stakeholder & Executive Communication

Professional Experience

GIS Coordinator

Nov 2017 – Present

Ikeja Electric • Lagos, Nigeria • Nigeria's largest electricity distributor

- Govern the multi-user enterprise geodatabase for a 3,800+ km distribution network serving 1M+ customers, setting and documenting data standards, automated validation rules, QA workflows, and CRS management across 23 spatial data layers.
- Prepare and maintain the electric utility network model and data that feed the advanced distribution management system (ADMS) on parts of the network, validating connectivity and topology so the ADMS model is correct; part of the ADMS implementation and rollout team.
- Gather user requirements and produce network diagrams (schematics) of the electrical distribution network.
- Administer a PostGIS-backed asset-management platform that is the single source of truth for 20,000+ assets, with role-based access control and single sign-on protecting sensitive operational data.
- Cut manual data-processing effort by 67% by engineering FME and Python (ArcPy, GeoPandas) ETL pipelines that integrate billing and field-capture data with automated validation.
- Reduced mean incident-response time by 40% with real-time outage dashboards; deliver the maps, dashboards, and reporting tools (ArcGIS Dashboards, Power BI) used by executives and planning teams.
- Accelerated field data collection by 45% across 8,500+ inspections by rolling out Esri Survey123, eliminating paper-based transcription errors at source.

GIS / Data Engineering Manager

Sep 2016 – Oct 2017

PoloSft Technologies • Lagos, Nigeria

- Led a team of 8 GIS professionals across concurrent project streams, delivering geospatial solutions on schedule for utility and infrastructure clients across West Africa.
- Designed cloud-based ETL pipelines for large-volume geospatial data processing and warehousing.

GIS / Database Specialist

Oct 2012 – Aug 2016

SpatialMatrix Ltd · Abeokuta, Ogun State, Nigeria

- Boosted database performance by 50% by optimizing complex spatial SQL queries and indexing strategies.
- Sustained 99.9% uptime across mission-critical production environments by migrating and administering enterprise PostgreSQL/PostGIS databases for oil & gas and infrastructure clients.
- Designed database architecture and delivered spatial analytics, dashboards, and cartography that turned raw geological and spatial data into decision-ready insight for oil & gas, NGO, and infrastructure clients.

Geologist

Jun 2010 – Aug 2012

Earth-Source Hydrocarbon · Nigeria

- Led geological field surveys and GPS-based data collection for hydrocarbon exploration, producing validated datasets that underpinned resource estimation and prospect appraisal.
- Integrated geological field data into ArcGIS for spatial analysis and mapping of prospective zones for exploration and drilling teams.

Across roles: designed geodatabase data models; worked with Microsoft SQL Server alongside PostgreSQL/PostGIS enterprise geodatabases; managed geodatabase versioning and replication; published data, geodata and OGC WFS services to ArcGIS Enterprise and ArcGIS Online; carried out software testing and release work for the GIS tools and APIs I build, writing test plans and running automated, functional and regression tests; administered PostGIS and AWS EC2 servers; and provided GIS user support and troubleshooting.

Earlier experience: GIS Technician, Spatial Technologies Limited, Nigeria: spatial ETL and QA, GPS/GNSS field campaigns and post-processing, geodatabase maintenance, data migration and topology validation, and network/proximity analysis for telecom and infrastructure projects.

Education

Master's Degree, Geographic Information Systems — University of Lagos, Nigeria

Bachelor of Technology (B.Tech), Geology — Federal University of Technology, Owerri, Nigeria

Certifications

- **GISP (Geographic Information Systems Professional)** — In progress (GISCI)
- **Python & SQL for Data Science; Tableau for Data Visualization** — Edureka, 2024
- **Big Data Analysis, Power Query, Power Pivot & VBA** — UrBizEdge, 2022

Technical Skills

Databases: Microsoft SQL Server, Oracle Spatial, PostgreSQL/PostGIS — enterprise geodatabase administration, versioning & replication, data modelling, database design, table partitioning, security & access control, performance tuning, migration

GIS & Geospatial: ArcGIS Pro, ArcMap, ArcGIS Enterprise, ArcGIS Online (publishing data & services, including geodata services), OGC services (WFS), ArcSDE, FME (Safe Software) ETL & format translation, QGIS, Survey123; topology & network connectivity validation; network diagrams (schematics); coordinate reference systems, reprojection, vertical datums & georeferencing; metadata & symbology; large raster processing (GeoTIFF/COG)

SQL & Scripting: Python (ArcPy, GeoPandas, PyQGIS, Pandas, Shapely, FastAPI), SQL, R — workflow automation, data validation, geoprocessing

Web Mapping: React, Next.js, TypeScript, Leaflet.js, versioned REST APIs returning JSON (Python/FastAPI) — internal web-map data products

Cloud & DevOps: AWS (S3, EC2, RDS), Docker, CI/CD (GitHub Actions), Git/GitHub; test plans, automated, functional & regression testing, release; server administration (PostGIS and AWS EC2 hosts)

AI / GeoAI: machine learning on spatial data, deep learning on imagery; LLMs, prompt engineering & generative AI; AI app/chatbot development; AI coding tools (GitHub Copilot, Cursor, Claude)

Reporting & BI: ArcGIS Dashboards, Power BI, Tableau, advanced Excel (Power Query, Power Pivot, VBA), Microsoft 365